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RECORDING DEVICE AND CUTTER UNIT

Abstract

A recording device includes a transport section that transports a medium in a transport direction along a transport path, a recording section that records an image on the medium transported by the transport section, a cutter unit including a cutter blade for processing the medium in the transport path, and a cutter carriage configured to move the cutter unit in a widthwise direction intersecting the transport direction, wherein the cutter unit is attachable to and detachable from the cutter carriage, and the cutter unit enters an exposed state in which the cutter blade becomes exposed in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, and enters a non-exposed state in which the cutter blade is not exposed in accordance with a detachment operation of detaching the cutter unit from the cutter carriage.

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Background/Summary

[0001] The present application is based on, and claims priority from JP Application Serial Number 2024-025333, filed Feb. 22, 2024, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

[0002] The present disclosure relates to a recording device and a cutter unit.

2. Related Art

[0003] For example, as described in JP-A-2013-86235, in a recording device that performs recording on a medium transported along a transport path, the medium is processed by a cutter blade. In such a recording device, a cutter holder that accommodates the cutter blade is configured to be attachable to and detachable from a cutter base that holds the cutter holder.

[0004] However, in such a recording device, the cutter blade is exposed. Therefore, it is desirable to improve safety with respect to the cutter blade when the cutter blade is replaced.

SUMMARY

[0005] A recording device that solves the above problem includes a transport section that transports a medium in a transport direction along a transport path; a recording section that records an image on the medium transported by the transport section; a cutter unit including a cutter blade for processing the medium in the transport path; and a cutter carriage configured to move the cutter unit in a widthwise direction intersecting the transport direction, wherein the cutter unit is attachable to and detachable from the cutter carriage and, in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, the cutter unit enters an exposed state in which the cutter blade becomes exposed and, in accordance with a detachment operation of detaching the cutter unit from the cutter carriage, enters a non-exposed state in which the cutter blade is not exposed. A cutter unit is attachable to and detachable from a cutter carriage with respect to a recording device, the recording device including a transport section that transports a medium in a transport direction along a transport path, a recording section that records an image on the medium transported by the transport section, and the cutter carriage, which moves the cutter unit in a widthwise direction intersecting the transport direction, wherein the cutter unit includes a cutter blade for processing the medium in the transport path, wherein the cutter unit enters an exposed state in which the cutter blade becomes exposed in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, and enters a non-exposed state non-exposed state in which the cutter blade is not exposed in accordance with a detachment operation of detaching the cutter unit from the cutter carriage.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. 1 is a perspective view showing a recording device of a first embodiment.

[0007] FIG. 2 is a perspective view showing an accommodation cassette of the first embodiment.

[0008] FIG. 3 is a perspective view showing the recording device of the first embodiment.

[0009] FIG. 4 is an explanatory diagram illustrating the recording device of the first embodiment.

[0010] FIG. **5** is a perspective view showing a cutting section of the first embodiment.
[0011] FIG. **6** is a perspective view showing the cutting section of the first embodiment.
[0012] FIG. **7** is a side view showing the cutting section of the first embodiment.
[0013] FIG. **8** is a top view showing a cutter unit of the first embodiment.
[0014] FIG. **9** is a cross-sectional view showing the cutter unit and the cutter carriage of the first embodiment.
[0015] FIG. **10** is a cross-sectional view showing the cutter unit and a cutter carriage of the first embodiment.
[0016] FIG. **11** is a cross-sectional view showing the cutter unit and the cutter carriage of the first embodiment.
[0017] FIG. **12** is a cross-sectional view showing the cutter unit and the cutter carriage of the first embodiment.
[0018] FIG. **13** is a perspective view showing a cutter unit according to a second embodiment.
[0019] FIG. **14** is a cross-sectional view showing the cutter unit of the second embodiment.
[0020] FIG. **15** is a cross-sectional view showing the cutter unit and a cutter carriage of the second embodiment.
[0021] FIG. **16** is a cross-sectional view showing the cutter unit and the cutter carriage of the second embodiment.
[0022] FIG. **17** is a perspective view showing a cutter unit of a third embodiment.
[0023] FIG. **18** is a top view showing the cutter unit and a cutter carriage of the third embodiment.
[0024] FIG. **19** is a cross-sectional view showing the cutter unit of the third embodiment.
[0025] FIG. **20** is a cross-sectional view showing the cutter unit and the cutter carriage of the third embodiment.
[0026] FIG. **21** is a cross-sectional view showing the cutter unit and the cutter carriage of the third embodiment.
[0027] FIG. **22** is a side view showing a cutter unit and a cutter carriage of a fourth embodiment.

DESCRIPTION OF EMBODIMENTS

First Embodiment

[0028] Hereinafter, an embodiment of a recording device will be described. In the following description, a direction intersecting the vertical direction Z is referred to as a widthwise direction X, and a direction intersecting the vertical direction Z and the widthwise direction X is referred to as a forward-rearward direction Y. One of the widthwise directions X is referred to as the first widthwise direction X1 and the other widthwise direction X is referred to as the second widthwise direction X2. One of the forward-rearward directions Y is referred to as the forward direction Y1 and the other forward-rearward direction Y is referred to as rearward direction Y2. Of the vertical directions Z, the upward direction is referred to as the upward direction Z1 and the downward direction is referred to as the downward direction Z2.

Configuration of Recording Device 10

[0029] As shown in FIG. **1**, a recording device **10** is a device that records an image on a medium **90** shown in FIG. **4**. The recording device **10** may be a device that records an image on the medium **90** by ejecting liquid. The liquid may be, for example, ink. The liquid may be, for example, in a plurality of colors or in one color. The medium **90** may be, for example, roll paper **91** and paper sheets **92** as illustrated in FIG. **4**. The roll paper **91** is a roll-shaped medium **90**. The paper sheets **92** are a sheet-like medium **90**. The paper sheets **92** are plain paper. The recording device **10** may be a device capable of performing recording on a plurality of types of medium **90**.

[0030] The recording device **10** includes a housing **11**. The housing **11** is the exterior of the recording device **10**. The housing **11** constitutes a front surface **12**, a rear surface **13**, a left side surface **14**, a right side surface **15**, a bottom surface **16**, and a top surface **17** of the recording device **10**, but is not limited to these. The housing **11** may constitute at least a part of the front surface **12**, a part of the rear surface **13**, a part of the left side surface **14**, a part of the right side surface **15**, a

part of the bottom surface **16**, and a part of the top surface **17** of the recording device **10**. The housing **11** accommodates various components that make up the recording device **10**.

[0031] The recording device **10** includes an operation panel **18**. The operation panel **18** may be provided on the front surface **12**. The operation panel **18** may be a touch panel. The operation panel **18** may include an operation section **18A** and a display section **18B**. The operation section **18A** can be operated by a user. The operation section **18A** may include a plurality of operation regions. The display section **18B** displays information related to the recording device **10**.

[0032] The recording device **10** includes a discharge tray **19**. The discharge tray **19** is a tray on which the medium **90** is placed after being recorded on and discharged. The discharge tray **19** is accessible from the forward direction **Y1**. The discharge tray **19** may be configured to be extendable in the forward-rearward direction **Y**.

[0033] The recording device **10** includes an accommodation cassette **20**. The accommodation cassette **20** constitutes a part of the front surface **12**. The accommodation cassette **20** is stored in the housing **11**. The accommodation cassette **20** is displaceable in the forward-rearward direction **Y**. The accommodation cassette **20** can be supplied with paper by being moved from the housing **11** in the forward direction **Y1**. The accommodation cassette **20** stores the medium **90**.

[0034] As shown in FIG. **2**, the accommodation cassette **20** includes a front section **21**, a cassette gripping section **22**, a first accommodation section **23A**, and a second accommodation section **23B**. The front section **21** is a wall section located in the forward direction **Y1**. The front section **21** constitutes a part of the front surface **12**. The cassette gripping section **22** is provided on the front section **21**. The cassette gripping section **22** can be gripped by a user.

[0035] The first accommodation section **23A** is configured to store the roll paper **91** as the medium **90**. The roll paper **91** corresponds to an example of a first medium. The second accommodation section **23B** is configured to accommodate the paper sheet **92** illustrated in FIG. **4** as the medium **90**. The paper sheet **92** corresponds to an example of a second medium. It is configured capable of feeding at least one of the roll paper **91** placed in the first accommodation section **23A** and the paper sheet **92** placed in the second accommodation section **23B**.

[0036] As shown in FIG. **3**, the housing **11** includes an opening section **11A** and an openable cover **11B**. The opening section **11A** is provided in the rear surface **13**. The opening section **11A** allows access to a cutting section **30**, which is accommodated in the housing **11**.

[0037] The openable cover **11B** is a cover that can open and close the opening section **11A**. The openable cover **11B** is pivotally attached in the downward direction **Z2** from the opening section **11A**, but is not limited thereto, and may be pivotally attached, for example, in the upward direction **Z1** from the opening section **11A**. In this way, the housing **11** enables access to the cutting section **30** via the opening section **11A** by opening the openable cover **11B**.

[0038] Specifically, for example, it is possible to resolve a jam of the medium **90** and perform maintenance on the cutting section **30**. Maintenance of the cutting section **30** includes replacement of a cutter unit **40** (to be described later).

Configuration of Replacement Unit Accommodation Section **39**

[0039] As shown in FIGS. **1** and **3**, the recording device **10** may include a replacement unit accommodation section **39**. The replacement unit accommodation section **39** is provided within the housing **11**. The replacement unit accommodation section **39** is, for example, a recessed space region formed in the housing **11**. The replacement unit accommodation section **39** can accommodate a replacement cutter unit **40**.

[0040] The recording device **10** may include a plurality of replacement unit accommodation sections **39**. The recording device **10** may include a first replacement unit accommodation section **39A** and a second replaceable unit accommodation section **39B** as the plurality of replacement unit accommodation sections **39**.

[0041] As shown in FIG. **1**, the first replacement unit accommodation section **39A** is provided on the front surface **12** side of the housing **11**. A replacement cutter unit **40** can be accommodated in

the first replacement unit accommodation section **39A**.

[0042] The housing **11** includes an accommodation cover **11C**. The accommodation cover **11C** constitutes the front surface **12** of the housing **11**. The accommodation cover **11C** is displaceable so as to open and close the first replacement unit accommodation section **39A**. The replacement cutter unit **40** accommodated in the first replacement unit accommodation section **39A** can be accessed from the forward direction **Y1** side in a state where the accommodation cover **11C** is opened.

[0043] As shown in FIG. **3**, the second replaceable unit accommodation section **39B** is provided on the rear surface **13** side of the housing **11**. A replacement cutter unit **40** can be accommodated in the second replaceable unit accommodation section **39B**.

[0044] The openable cover **11B** can be displaced so as to open and close the second replaceable unit accommodation section **39B**. The replacement cutter unit **40** that is accommodated in the second replaceable unit accommodation section **39B** can be accessed from the rearward direction **Y2** side while the openable cover **11B** is opened.

Configuration of Feeding Section **31**

[0045] As illustrated in FIG. **4**, the recording device **10** includes the cutting section **30**, a feeding section **31**, a transport section **32**, a recording section **33**, a medium support section **34**, and a control section **35**. When the accommodation cassette **20** is mounted in the housing **11**, the feeding section **31** is positioned in the upward direction **Z1** from the accommodation cassette **20**. The feeding section **31** is configured to feed the medium **90** placed on the accommodation cassette **20**.

[0046] The feeding section **31** includes a feed roller **31A** and a roller support section **31B**. The feed roller **31A** is a roller for feeding the medium **90**. The roller support section **31B** is pivotally mounted at its base end section so as to be rotatable about a rotation shaft **31C**. The rotation shaft **31C** extends in the widthwise direction **X**. The roller support section **31B** is configured to rotatably support the feed roller **31A** at the tip end section thereof. The roller support section **31B** is inclined so that the feed roller **31A** abuts the medium **90**.

[0047] The accommodation cassette **20** includes a feed path **20A** for the roll paper **91**. The feed path **20A** is positioned in the downward direction **Z2** from the first accommodation section **23A** and the second accommodation section **23B**. The feed path **20A** is a path that connects the first accommodation section **23A** and a transport path **24** (to be described later).

[0048] The accommodation cassette **20** includes a feed flap **20B**. The feed flap **20B** is provided on a support plate of the second accommodation section **23B**. The feed flap **20B** is displaced to a paper sheet position and to a roll paper position. The feed flap **20B** may be displaced by rotation or by sliding.

[0049] The paper sheet position is the position when recording is to be performed on the paper sheets **92** accommodated in the second accommodation section **23B**. The paper sheet position is the position at which the paper sheets **92** stored in the second accommodation section **23B** are supported by closing an opening or a notch provided in the bottom plate of the second accommodation section **23B**.

[0050] The roll paper position is the position when recording is to be performed on the roll paper **91** accommodated in the first accommodation section **23A**. The roll paper position is a position at which the feed roller **31A** is brought into contact with the roll paper **91** in the feed path **20A** by opening the opening or the notch provided in the bottom plate of the second accommodation section **23B**.

[0051] When a paper sheet **92** is stored in the second accommodation section **23B**, the feed roller **31A** comes into contact with the paper sheet **92** due to the inclination of the roller support section **31B**. More specifically, the feed roller **31A** contacts the uppermost paper sheet **92** stored in the second accommodation section **23B**, that is, the paper sheet **92** placed furthest in the upward direction **Z1**. The feed roller **31A** rotates in the feed direction while in contact with the paper sheets **92**, thereby feeding out paper sheets **92** one sheet at a time. In this case, the feed flap **20B** is arranged in the paper sheet position. Therefore, the second accommodation section **23B** can stably

store the paper sheets **92**.

[0052] When the feed flap **20B** is positioned at the roll paper position and no paper sheets **92** are stored in the second accommodation section **23B**, the feed roller **31A** is brought into contact with the roll paper **91** positioned in the feed path **20A** due to the inclination of the roller support section **31B**. The feed roller **31A** rotates in the feed direction while in contact with the roll paper **91** in the feed path **20A** to feed the roll paper **91** that the feed roller **31A** contacts.

Configuration of Transport Section **32**

[0053] The transport section **32** is configured to transport the medium **90**. The transport section **32** transports the medium **90** that was fed out by the feeding section **31**, in the transport direction **D** along the transport path **24**. The transport path **24** is a path for transporting the medium **90**.

[0054] The transport path **24** may include a path for transporting the roll paper **91** and a path for transporting the paper sheet **92**. The transport path **24** may include an inversion path **24A** for turning the medium **90** over. The inversion path **24A** is a path for transporting the medium **90** that was inverted by an inversion section **28** (to be described later).

[0055] The transport section **32** includes a plurality of rollers and a plurality of motors serving as drive sources. In particular, the transport section **32** includes a transport roller pair **25**. The transport roller pair **25** is located upstream of the medium support section **34** in the transport path **24**. The transport roller pair **25** transports the medium **90** along the transport path **24** to the medium support section **34**.

[0056] The transport section **32** includes a first discharge roller pair **26** and a second discharge roller pair **27**. The first discharge roller pair **26** and the second discharge roller pair **27** are located downstream of the medium support section **34** in the transport path **24**. The second discharge roller pair **27** is positioned downstream of the first discharge roller pair **26** in the transport path **24**. The first discharge roller pair **26** and the second discharge roller pair **27** discharge the medium **90** after recording along the transport path **24**.

[0057] The transport section **32** may include the inversion section **28**. That is, the recording device **10** includes the inversion section **28**. The inversion section **28** is configured to invert the medium **90** after recording. The second discharge roller pair **27** rotates in a direction opposite to the rotation direction in which the medium **90** is discharged, thereby transporting the medium **90** along the inversion path **24A** to the inversion section **28**. The inversion section **28** is configured to invert the printed-on medium **90** that was transported to the second discharge roller pair **27**.

[0058] The inversion section **28** includes an inversion flap **29**. The inversion flap **29** is provided at a position along the inversion path **24A**. The inversion flap **29** switches the direction in which the medium **90** is transported. In particular, the inversion flap **29** switches whether or not to cause the medium **90** to travel along the inversion path **24A**. In this manner, the inversion section **28** causes the recorded-on and inverted medium **90** to enter the transport path **24** further upstream than the medium support section **34**.

[0059] By this, the medium **90** with an image recorded on the front surface of the medium **90** is transported to the medium support section **34** in a state of having been inverted by the inversion section **28**. By this, an image can also be recorded on the rear surface of the medium **90** by the recording section **33**. In this way, the recording section **33** can perform recording on both surfaces of the medium **90**. That is, recording section **33** can perform double-sided printing.

Configuration of Recording Section **33**

[0060] The recording section **33** is configured to record an image on the medium **90**. The recording section **33** may be configured to record an image on the medium **90** by ejecting liquid onto the medium **90**. The recording section **33** is positioned in the upward direction **Z1** from the medium support section **34**.

[0061] The recording section **33** includes a recording carriage **36** and a liquid ejection head **37**. The recording carriage **36** supports the liquid ejection head **37** from the upward direction **Z1**. The liquid ejection head **37** is provided in the downward direction **Z2** portion from the recording carriage **36**.

[0062] The liquid ejection head **37** includes a nozzle surface **38**. The nozzle surface **38** is a surface facing in the downward direction **Z2**. The nozzle surface **38** is a surface facing the medium support section **34**. The nozzle surface **38** is a surface in which a plurality of nozzles (not shown) are opened. Each of the plurality of nozzles opens toward the medium **90** supported by the medium support section **34**. Each of the plurality of nozzles discharges liquid.

[0063] The liquid ejection head **37** ejects liquid from the plurality of nozzles toward the medium **90** supported by the medium support section **34**. In this way, the recording section **33** is configured to perform recording on the medium **90** by ejecting liquid onto the medium **90**. The liquid ejection head **37** is a serial head type which ejects liquid in accordance with the movement of the recording carriage **36** in the widthwise direction **X**, but it may also be a line head type.

[0064] The medium support section **34** is configured to support the medium **90**. In particular, the medium support section **34** supports the medium **90** in a recording region in which recording is performed on the medium **90** by the recording section **33**. That is, the medium support section **34** supports the medium **90** on which recording is performed by the recording section **33**.

Configuration of Control Section 35

[0065] The control section **35** controls the recording device **10**. The control section **35** controls each unit in the recording device **10**. The control section **35** controls various operations performed by the recording device **10**. The control section **35** may be communicably connected to a terminal device (not shown).

[0066] The control section **35** can be configured as a circuit including α : one or more processors that execute various processes in accordance with a computer program, β : one or more dedicated hardware circuits that execute at least some of the various processes, or γ : a combination of these. The hardware circuit is, for example, an application specific integrated circuit. The processor includes a CPU and memory, such as RAM and ROM, which stores program code or instructions configured to cause the CPU to perform processes. Memory, or computer-readable medium, includes any readable medium that can be accessed by a general purpose or special purpose computer.

Configuration of Cutting Section 30

[0067] The cutting section **30** is configured to cut the medium **90**. The cutting section **30** is provided at the rear surface **13** side of the recording device **10**. The cutting section **30** is provided at a position along the transport path **24**. The cutting section **30** is configured to cut the roll paper **91** transported along the transport path **24** in the widthwise direction **X**.

[0068] The cutting section **30** is provided at a position along the transport path **24** between the feeding section **31** and the recording section **33**, but may also be provided at a position along the transport path **24** between the recording section **33** and the discharge tray **19**.

[0069] The cutting section **30** is located at the forward direction **Y1** side of the opening section **11A**. The cutting section **30** is exposed through the opening section **11A** when the openable cover **11B** is open. By this, the cutting section **30** can be accessed by the user via the opening section **11A** when the openable cover **11B** is open.

[0070] As shown in FIG. 5, the cutting section **30** includes the cutter unit **40**, a cutter carriage **41**, a carriage support section **42**, a fixed blade **43**, and a carriage movement section **44**. That is, the recording device **10** includes the cutter unit **40**, the cutter carriage **41**, the carriage support section **42**, the fixed blade **43**, and the carriage movement section **44**.

[0071] The cutter unit **40** is a unit for cutting the medium **90**. The cutter unit **40** is mounted on the cutter carriage **41**. The cutter unit **40** is attachable to and detachable from the cutter carriage **41**. In other words, the cutter carriage **41** can have the cutter unit **40** attached to it and detached from it.

[0072] The cutting section **30** is arranged so that the cutter blade **50** protrudes along an advance/retract direction **A**. The direction in which the cutter blade **50** protrudes in the advance/retract direction **A** is referred to as the advance direction **A1**, and a direction opposite to the advance direction **A1** is referred to as the retraction direction **A2**. The advance/retract direction

A is a direction intersecting the transport direction D and the widthwise direction X. The advance/retract direction A is a direction inclined upward toward forward direction Y1. A direction intersecting the advance/retract direction A and the widthwise direction X is referred to as an intersecting direction B. A direction toward the upward direction Z1 side in the intersecting direction B is referred to as the first intersecting direction B1, and a direction opposite to the first intersecting direction B1 is referred to as the second intersecting direction B2. The cutting section 30 may be disposed such that the cutter blade 50 protrudes in the forward direction Y1. In this case, the advance/retract direction A coincides with the forward-rearward direction Y and the intersecting direction B coincides with the vertical direction Z.

[0073] In a state of being supported by the carriage support section 42, the cutter carriage 41 is movable along the widthwise direction X by driving force from the carriage movement section 44. In a state in which the cutter unit 40 is mounted, the cutter carriage 41 moves the cutter unit 40 in the widthwise direction X, which intersects the transport direction D.

[0074] At least the cutter unit 40 and the cutter carriage 41 are accommodated in the housing 11. That is, the housing 11 accommodates at least the cutter unit 40 and the cutter carriage 41. At least the cutter unit 40 and the cutter carriage 41 are accessible through the opening section 11A. That is, the housing 11 enables, via the opening section 11A, a detachment operation of detaching the cutter unit 40 from the cutter carriage 41 and an attachment operation of attaching the cutter unit 40 to the cutter carriage 41.

[0075] The carriage support section 42 extends in the widthwise direction X. The carriage support section 42 is configured to support the cutter carriage 41. The carriage support section 42 guides the cutter carriage 41 so as to be movable along the widthwise direction X.

[0076] The fixed blade 43 extends in the widthwise direction X. The fixed blade 43 may be configured in an L shape as viewed from the first widthwise direction X1. The fixed blade 43 is provided at a position facing the cutter blade 50 of the cutter unit 40 with the transport path 24 interposed therebetween. In this manner, the cutter blade 50 and the fixed blade 43 cut the medium 90 transported along the transport path 24.

[0077] The carriage movement section 44 includes a pair of pulleys 45, a timing belt 46, and a carriage drive section 47. The pair of pulleys 45 is provided at both end sections of the carriage support section 42 in the widthwise direction X. The pair of pulleys 45 includes a first pulley (not shown) and a second pulley 45B. The first pulley is provided at the end section of the carriage support section 42 in the first widthwise direction X1. The second pulley 45B is provided at the end section of the carriage support section 42 in the second widthwise direction X2.

[0078] The timing belt 46 is wound around the pair of pulleys 45. The timing belt 46 may be an endless belt. The timing belt 46 is connected to the cutter carriage 41.

[0079] The carriage drive section 47 is a drive source for moving the cutter carriage 41 in the widthwise direction X. The carriage drive section 47 may be a motor. The carriage drive section 47 may be provided at the end section of the carriage support section 42 in the second widthwise direction X2. The carriage drive section 47 may rotate the second pulley 45B via a transmission member (not shown). By this, the carriage drive section 47 moves the cutter carriage 41 in the widthwise direction X.

[0080] The cutting section 30 may include a position detection section (not shown). The position detection section detects the position of the cutter carriage 41 in the widthwise direction X. The position detection section may be an encoder that detects the position of the cutter carriage 41 by detecting the amount of rotation of at least one of the pair of pulleys 45. The position detection section may be an encoder that detects the position of the cutter carriage 41 by detecting position information of the cutter carriage 41 in the widthwise direction X. The position detection section outputs a signal indicating a detection result to control section 35.

Configuration of Cutter Unit 40

[0081] As shown in FIGS. 6 and 7, the cutter unit 40 includes the cutter blade 50 and a cutter

accommodation section **51**. The cutter blade **50** is a member for cutting the medium **90** in the transport path **24**. The cutter accommodation section **51** can accommodate the cutter blade **50**. [0082] The cutter accommodation section **51** includes a positioning section **52**. The positioning section **52** is a member for positioning the cutter unit **40** with respect to the cutter carriage **41**. The positioning section **52** is configured to protrude in the first intersecting direction **B1**. The positioning section **52** may have a plate shape extending along the advance/retract direction **A** and the intersecting direction **B**. The positioning section **52** is also a part for gripping the cutter unit **40**. Configuration of Cutter Carriage **41**

[0083] As shown in FIGS. **5** and **6**, the cutter carriage **41** is movable along the widthwise direction **X** from a standby position **P11**. The standby position **P11** is located on the first widthwise direction **X1** side. The standby position **P11** is a home position in which the cutter carriage **41** waits while the cutting section **30** is not operating.

[0084] The cutter carriage **41** includes a unit holding section **60** and a unit fixing section **61**. The unit holding section **60** holds the cutter unit **40**. The unit holding section **60** is movable along the widthwise direction **X** by driving force from the carriage movement section **44**.

[0085] As shown in FIG. **6**, the unit holding section **60** includes a unit support section **63**, a pair of unit guide sections **64**, and an inclined section **65**. The unit support section **63** is configured to support the cutter unit **40** from the second intersecting direction **B2** when the cutter unit **40** is mounted on the cutter carriage **41**. The upper surface of the unit support section **63** is a flat surface. The unit support section **63** may have a flat plate shape.

[0086] The pair of unit guide sections **64** are provided at both end sections of the unit support section **63** in the widthwise direction **X**. The unit guide section **64** is a wall section that protrudes in the first intersecting direction **B1**. The pair of unit guide sections **64** guide the cutter unit **40** to be movable along the advance/retract direction **A**.

[0087] When the cutter unit **40** is mounted on the cutter carriage **41**, the cutter unit **40** is held in the holding region **60A**. The holding region **60A** is a region surrounded by the unit support section **63** and the pair of unit guide sections **64**. As described above, the holding region **60A** is a region where the cutter unit **40** is held when the cutter unit **40** is mounted on the cutter carriage **41**. That is, the holding region **60A** is a mounting region in which the cutter unit **40** is mounted.

[0088] The unit holding section **60** is open in the advance direction **A1** of the holding region **60A**. Therefore, when the cutter unit **40** is mounted on the cutter carriage **41**, the cutter blade **50** is such that the cutter blade **50** can be exposed toward the transport path **24**.

[0089] The inclined section **65** is provided at a retraction direction **A2** portion of the unit support section **63**. The inclined section **65** is configured to be inclined with respect to the unit support section **63**. The inclined section **65** is inclined downward toward the advance direction **A1**. The inclined section **65** is configured so that when the cutter unit **40** is detached from the cutter carriage **41**, the inclined section **65** comes into contact with the end section of the cutter unit **40** in the retraction direction **A2**.

[0090] Together with the unit holding section **60**, the unit fixing section **61** fixes the cutter unit **40** in place by contacting the cutter unit **40**, which is held by the unit holding section **60**, from the first intersecting direction **B1**. That is, the unit fixing section **61** fixes the cutter unit **40** in place when the cutter unit **40** is mounted in the cutter carriage **41**.

[0091] The unit fixing section **61** includes a main body section **61A**, a pair of arm sections **61B**, and a gripping section **61C**. The main body section **61A** has a flat plate shape. The main body section **61A** is a part for fixing the cutter unit **40** in place by contacting the cutter unit **40** held by the unit holding section **60** from the first intersecting direction **B1**.

[0092] The arm sections **61B** are provided at the advance direction **A1** portion of the main body section **61A**. The pair of arm sections **61B** are aligned in the widthwise direction **X**. An opening is provided at the tip end section of the arm section **61B**, and a coupling shaft **62** provided on the main body section **61A** is inserted into the opening. The coupling shaft **62** is a shaft that protrudes

outward along the widthwise direction X from the unit holding section **60**. By this, the unit fixing section **61** is rotatable with respect to the unit holding section **60** about an axis of rotation along the widthwise direction X.

[0093] As shown in FIGS. 5 and 6, the unit fixing section **61** is movable between a fixed position **P21** and a release position **P22**. That is, the unit fixing section **61** is rotatable between the fixed position **P21** and the release position **P22** around the coupling shaft **62** in a state in which the cutter unit **40** is mounted.

[0094] The fixed position **P21** is a position at which the cutter unit **40** held by the unit holding section **60** is fixed. The fixed position **P21** is a position at which the unit fixing section **61** contacts the cutter unit **40** held by the unit holding section **60** from the first intersecting direction **B1** side.

[0095] The release position **P22** is a position in which the cutter unit **40** held by the unit holding section **60** is not fixed and the fixing of the cutter unit **40** is released. The release position **P22** is a position in which the unit fixing section **61** does not contact, from the first intersecting direction **B1**, the cutter unit **40** that is held by the unit holding section **60**. The release position **P22** is a position at which the cutter unit **40** held by the unit holding section **60** can be attached and detached.

[0096] The gripping section **61C** is provided at the retraction direction **A2a** side of the main body section **61A**. The gripping section **61C** is a part that can be gripped by a user when the unit fixing section **61** is to be moved. In this way, the unit fixing section **61** is movable by being gripped by the user.

[0097] The main body section **61A** is provided with a positioning opening section **61D**. The positioning opening section **61D** extends along the advance/retract direction **A**. While the cutter unit **40** is held by the unit holding section **60**, the positioning section **52** enters into and is retracted from the positioning opening section **61D** by movement of the unit fixing section **61** between the fixed position **P21** and the release position **P22**.

[0098] When the unit fixing section **61** is disposed at the fixed position **P21** while the cutter unit **40** is held by the unit holding section **60**, the positioning section **52** is inserted into the positioning opening section **61D**. By this, when the cutter unit **40** is mounted on the cutter carriage **41**, the cutter unit **40** is positioned on the cutter carriage **41**.

[0099] As shown in FIG. 6, the main body section **61A** includes a protrusion section **61E**. The main body section **61A** may include a pair of protrusion sections **61E**. The pair of protrusion sections **61E** are provided at both widthwise direction X end sections of the main body section **61A**. The protrusion sections **61E** have an arc shape in the advance direction **A1**. The protrusion sections **61E** are configured to protrude in the second intersecting direction **B2** when the unit fixing section **61** is disposed at the fixed position **P21**.

[0100] The protrusion sections **61E** come into contact with the cutter unit **40** when the cutter unit **40** is mounted on the cutter carriage **41**. By this, the protrusion section **61E** fixes the cutter unit **40** to the cutter carriage **41**.

[0101] The attachment operation of attaching the cutter unit **40** to the cutter carriage **41** includes an operation of disposing the cutter unit **40** on the cutter carriage **41** and an operation of moving the unit fixing section **61** to the fixed position **P21**. The detachment operation of detaching the cutter unit **40** from the cutter carriage **41** includes an operation of moving the unit fixing section **61** to the release position **P22** and an operation of taking the cutter unit **40** out from the cutter carriage **41**.

Configuration of Cutter Accommodation Section **51**

[0102] As shown in FIGS. 8 and 9, the cutter accommodation section **51** includes a cutter case **53**, a base section **56**, a fixing member **57**, and biasing sections **58**. That is, the cutter unit **40** includes the cutter case **53**, the base section **56**, the fixing member **57**, and the biasing sections **58**. The cutter accommodation section **51** may include a pair of biasing sections **58**.

[0103] The cutter case **53** is configured to accommodate the cutter blade **50**, the base section **56**, the fixing member **57**, and the biasing sections **58**. The cutter case **53** includes a case **54** and a

cover 55. That is, the cutter unit 40 includes the case 54 and the cover 55.

[0104] The case 54 is configured to support the base section 56 from the second intersecting direction B2 side. The case 54 is configured to accommodate the cutter blade 50, the base section 56, the fixing member 57, and the biasing sections 58.

[0105] The case 54 includes a base guide section 54A. The base guide section 54A is a wall section provided so as to protrude from the case 54 in the first intersecting direction B1. The base guide section 54A is provided at both end sections of the case 54 in the widthwise direction X and at the end section of the case 54 in the retraction direction A2. The base guide section 54A guides the base section 56 to move along the advance/retract direction A. The base guide section 54A constitutes the end section 54B of the case 54 in the retraction direction A2.

[0106] The cover 55 is provided on the first intersecting direction B1 portion of the case 54. The cover 55 is fixed to the case 54. The cover 55 is configured to cover the base section 56 from the first intersecting direction B1. The cover 55 is configured to accommodate the cutter blade 50, the base section 56, the fixing member 57, and the biasing sections 58 together with the case 54. The cover 55 includes the positioning section 52.

[0107] The cover 55 includes a first opening section 55A. The cover 55 may include a pair of first opening sections 55A. The pair of first opening sections 55A is arranged in the widthwise direction X. The first opening section 55A is provided so as to penetrate the cover 55 in the intersecting direction B. The first opening section 55A is provided at the retraction direction A2 side of the cover 55.

[0108] The first opening section 55A may be an elongated hole extending along the advance/retract direction A. A contact section 56C (to be described later) is inserted into the first opening section 55A. The first opening section 55A may have a shape that enables the contact section 56C to move in the advance/retract direction A. The biasing sections 58 may be held in the first opening sections 55A along the advance/retract direction A.

[0109] The first opening section 55A is provided so that the protrusion section 61E can be inserted into it and removed from it. When the unit fixing section 61 is disposed in the fixed position P21, the protrusion section 61E is inserted into the first opening section 55A. By this, the unit fixing section 61 fixes the cutter unit 40 to the cutter carriage 41.

[0110] When the unit fixing section 61 is disposed at the release position P22, the protrusion section 61E separates from the first opening section 55A. By this, the unit fixing section 61 releases the fixed state of the cutter unit 40 to the cutter carriage 41.

[0111] The cover 55 includes a second opening section 55B. The second opening section 55B is provided in the advance direction A1 side of the cover 55. The second opening section 55B is provided so as to penetrate the cover 55 along the intersecting direction B.

[0112] The second opening section 55B may be an elongated hole extending along the advance/retract direction A. The cutter holding section 56B (to be described later) is inserted into the second opening section 55B. The second opening section 55B may have a shape that enables the cutter holding section 56B to move in the advance/retract direction A.

[0113] The base section 56 is configured to hold the cutter blade 50. The base section 56 is movable in the cutter case 53 along the advance/retract direction A. In this way, the cutter blade 50 held by the base section 56 moves along the advance/retract direction A by the base section 56 moving along the advance/retract direction A.

[0114] As shown in FIGS. 10 to 12, while accommodated in the cutter case 53, the base section 56 is movable between an exposed position P31 and a non-exposed position P32 along the advance/retract direction A. The exposed position P31 is a position at which the cutter blade 50 is exposed. The non-exposed position P32 is a position in which the cutter blade 50 is not exposed.

[0115] As shown in FIGS. 8 and 9, the base section 56 includes a base body section 56A. The base body section 56A is configured to support the cutter blade 50 from the second intersecting direction B2. The base body section 56A has a flat plate shape. The base body section 56A is movable along

the advance/retract direction A by being guided by the base guide section 54A. That is, the base section 56 is movable in the advance/retract direction A while accommodated in the cutter case 53. [0116] The base section 56 includes a cutter holding section 56B. The cutter holding section 56B is configured to protrude from the base body section 56A toward the first intersecting direction B1. The cutter holding section 56B is configured to hold the cutter blade 50. The cutter holding section 56B may be a shaft that holds the cutter blade 50.

[0117] The fixing member 57 is mounted on the cutter holding section 56B. The fixing member 57 fixes the cutter blade 50 to the base body section 56A by contacting the cutter blade 50 from the first intersecting direction B1. By this, the cutter blade 50 is fixed between the base body section 56A and the fixing member 57 while being held by the cutter holding section 56B.

[0118] The cutter holding section 56B is provided so as to be inserted into the second opening section 55B. When the base section 56 moves along the advance/retract direction A, the cutter holding section 56B moves along the second opening section 55B.

[0119] The base section 56 includes a contact section 56C. The base section 56 may include a pair of contact sections 56C. The pair of contact sections 56C are provided so as to be aligned in the widthwise direction X. The contact section 56C is provided at the retraction direction A2 side of the base body section 56A. The contact section 56C is configured to protrude in the first intersecting direction B1.

[0120] The contact section 56C is provided so as to be inserted into the first opening section 55A. The contact section 56C is configured to not to protrude from the front face of the cutter unit 40 in a state of being inserted into the first opening section 55A when the cutter unit 40 is mounted on the cutter carriage 41. The contact section 56C comes into contact with the cutter carriage 41 in accordance with an attachment operation of attaching the cutter unit 40 on the cutter carriage 41.

[0121] The pair of biasing sections 58 are provided so as to be aligned in the widthwise direction X. The biasing sections 58 are provided between the cutter case 53 and the base section 56. The biasing sections 58 may be provided between the case 54 and the contact section 56C. The biasing sections 58 may be provided so as to be accommodated in the first opening section 55A. The biasing section 58 is configured to bias the base section 56 in the retraction direction A2 with respect to the cutter case 53. The biasing section 58 may be a compression spring.

Operation of the First Embodiment

[0122] The operation of the first embodiment will be described.

[0123] As shown in FIG. 10, when the unit fixing section 61 is at the release position P22, the attachment operation of the cutter unit 40 to the cutter carriage 41 is performed. Specifically, when the unit fixing section 61 is at the release position P22, the cutter unit 40 is disposed in the holding region 60A of the cutter carriage 41 from the retraction direction A2 side.

[0124] In such a case, the protrusion section 61E is not inserted into the first opening section 55A and does not contact the contact section 56C. By this, the cutter unit 40 is disposed at the non-exposed position P32 by the biasing force of the biasing section 58.

[0125] As shown in FIG. 11, when the unit fixing section 61 rotates from the release position P22 toward the fixed position P21, the protrusion section 61E is inserted into the first opening section 55A. By this, the protrusion section 61E comes into contact with the contact section 56C. That is, the contact section 56C comes into contact with the cutter carriage 41 in accordance with the attachment operation of the cutter unit 40 to the cutter carriage 41.

[0126] As shown in FIG. 12, when the unit fixing section 61 rotates from the release position P22 to the fixed position P21, the protrusion section 61E presses against the contact section 56C. In such a case, the base section 56 moves in the advance direction A1 against the biasing force of the biasing section 58. By this, the cutter unit 40 moves from the non-exposed position P32 to the exposed position P31 so that the cutter blade 50 held by the base section 56 is exposed from the cutter accommodation section 51.

[0127] As described above, in accordance with the attachment operation of the cutter unit 40 onto

the cutter carriage **41**, the cutter carriage **41** comes into contact with the contact section **56C** and, by this, the base section **56** moves from the non-exposed position **P32** to the exposed position **P31** against the biasing force of the biasing section **58**. By this, the cutter unit **40** enters an exposed state in which the cutter blade **50** is exposed from the cutter case **53**. In this way, the cutter unit **40** is brought into the exposed state in accordance with the attachment operation to the cutter carriage **41**. The exposed state is a state in which the cutter blade **50** is exposed.

[0128] Next, a detachment operation of detaching the cutter unit **40** from the cutter carriage **41** is performed when the unit fixing section **61** is at the fixed position **P21**. Specifically, the unit fixing section **61** rotates from the fixed position **P21** toward the release position **P22** in accordance with a detachment operation of detaching the cutter unit **40** from the cutter carriage **41**.

[0129] In such a case, as shown in FIG. **10**, the protrusion section **61E** retreats from the first opening section **55A**, so that the protrusion section **61E** does not contact the contact section **56C**. By this, the cutter unit **40** moves from the exposed position **P31** to the non-exposed position **P32** by the biasing force of the biasing section **58** and, by this, the cutter blade **50** held by the base section **56** is not exposed from the cutter accommodation section **51**.

[0130] As described above, in accordance with the detachment operation of detaching the cutter unit **40** from the cutter carriage **41**, the cutter carriage **41** and the contact section **56C** do not contact each other. In this case, the base section **56** moves from the exposed position **P31** to the non-exposed position **P32** under the biasing force of the biasing section **58**. By this, the cutter unit **40** is in the non-exposed state in which the cutter blade **50** is not exposed from the cutter case **53**. In this way, the cutter unit **40** is brought into the non-exposed state in accordance with the detachment operation of detaching it from the cutter carriage **41**. The non-exposed state is a state where the cutter blade **50** is not exposed.

Effects of the First Embodiment

[0131] Effects of the first embodiment will be described.

[0132] (1-1) The cutter blade **50** enters the exposed state, in which the cutter blade **50** is exposed, in accordance with the attachment operation of attaching the cutter unit **40** to the cutter carriage **41**. The cutter blade **50** enters the non-exposed state, in which the cutter blade **50** is not exposed, in accordance with the detachment operation of detaching the cutter unit **40** from the cutter carriage **41**. According to this configuration, it is possible to process the medium **90** by exposing the cutter blade **50** in accordance with the attachment operation of attaching the cutter unit **40** to the cutter carriage **41**.

[0133] On the other hand, in accordance with the detachment operation of the cutter unit **40** from the cutter carriage **41**, the cutter blade **50** is not exposed. Therefore, it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

[0134] (1-2) The protrusion section **61E** and the contact section **56C** come into contact with each other in accordance with the attachment operation of attaching the cutter unit **40** to the cutter carriage **41**. In this case, the exposed state is achieved by moving the base section **56** to the exposed position **P31** against the biasing force of the biasing section **58**. By this, the medium **90** can be processed by exposing the cutter blade **50**.

[0135] On the other hand, in accordance with the detachment operation of detaching the cutter unit **40** from the cutter carriage **41**, the protrusion section **61E** and the contact section **56C** do not contact each other. In this case, the base section **56** is moved to the non-exposed position **P32** by the biasing force of the biasing section **58**, so that the non-exposed state is established. Therefore, it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

[0136] (1-3) When the cutter unit **40** is mounted on the cutter carriage **41**, the contact section **56C** does not protrude from the front face of the cutter unit **40**. By this, an erroneous operation can be suppressed. Therefore, it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

[0137] (1-4) The replacement unit accommodation section **39** can accommodate the replacement

cutter unit **40**. According to this configuration, it is possible to easily replace the cutter unit **40** by taking out the replacement cutter unit **40** out of the replacement unit accommodation section **39**.

Second Embodiment

[0138] Next, a second embodiment will be described. In the following description, redundant descriptions of configurations identical to those of the previously described embodiment will be omitted or simplified, and configurations that differ from the previously described embodiment will be detailed.

[0139] As shown in FIGS. **13** to **16**, in the second embodiment, the cover **55** includes a displacement section **55C**. That is, the cutter unit **40** includes the displacement section **55C**. The displacement section **55C** protrudes from the bottom surface of the cover **55** in the second intersecting direction **B2**. The displacement section **55C** is positioned on the retraction direction **A2** side of the cover **55**. The case **54** and the cover **55** are not fixed to each other. The cover **55** and the base section **56** are fixed to each other. Therefore, the cover **55** and the base section **56** move in conjunction with each other. That is, the displacement section **55C** is displaced in conjunction with the base section **56**.

[0140] The cover **55** and the base section **56** integrally move in the advance/retract direction **A**. That is, the cutter holding section **56B** and the contact section **56C** are not movable relative to the cover **55**. The first opening section **55A** may not have an elongated hole shape. The second opening section **55B** may not necessarily have an elongated hole shape.

[0141] The base section **56** is movable between the exposed position **P31** and the non-exposed position **P32** in a state of being accommodated in the case **54**. The exposed position **P31** is a position at which the cutter blade **50** is exposed. The non-exposed position **P32** is a position in which the cutter blade **50** is not exposed.

[0142] The case **54** includes a locking section **54C**. The locking section **54C** is configured to protrude in the first intersecting direction **B1** the upper surface of the case **54**. In conjunction with the base section **56**, the displacement section **55C** moves along the advance/retract direction **A** over the locking section **54C**. The locking section **54C** locks with the displacement section **55C** in the exposed position **P31** and in the non-exposed position **P32**.

[0143] As described above, the exposed position **P31** is a position in which the displacement section **55C** is arranged at the advance direction **A1** side of the locking section **54C**. The non-exposed position **P32** is a position in which the displacement section **55C** is arranged at the retraction direction **A2** side of the locking section **54C**. The cutter unit **40** may not include the biasing section **58**.

Operation of the Second Embodiment

[0144] The operation of the second embodiment will be described.

[0145] As shown in FIG. **15**, in accordance with an attachment operation of attaching the cutter unit **40** to the cutter carriage **41**, the unit fixing section **61** rotates from the release position **P22** toward the fixed position **P21**, whereupon the protrusion section **61E** comes into contact with the contact section **56C**. That is, the contact section **56C** comes into contact with the cutter carriage **41** in accordance with the attachment operation of the cutter unit **40** to the cutter carriage **41**.

[0146] When the unit fixing section **61** rotates from the release position **P22** to the fixed position **P21**, the protrusion section **61E** presses the contact section **56C**, resulting in the cover **55** and the base section **56** moving in the advance direction **A1**.

[0147] In this case, the displacement section **55C** moves over the locking section **54C** toward the advance direction **A1** and moves to the advance direction **A1** side of the locking section **54C**. By this, the cutter blade **50** held by the base section **56** is exposed from the cutter accommodation section **51**. In this way, the cutter unit **40** moves from the non-exposed position **P32** to the exposed position **P31**.

[0148] As described above, in accordance with an attachment operation of attaching the cutter unit **40** to the cutter carriage **41**, the cutter carriage **41** and the contact section **56C** come into contact

with each other, so that the displacement section 55C moves over the locking section 54C and the base section 56 becomes disposed at the exposed position P31. By this, the cutter unit 40 is in an exposed state in which the cutter blade 50 is exposed from the case 54.

[0149] Next, as shown in FIG. 16, a detachment operation of detaching the cutter unit 40 from the cutter carriage 41 is performed. Specifically, the unit fixing section 61 rotates from the fixed position P21 to the release position P22 in accordance with the detachment operation of detaching the cutter unit 40 from the cutter carriage 41.

[0150] When the unit fixing section 61 is at the release position P22, the cutter unit 40 is removed from the cutter carriage 41. In this case, when the cutter unit 40 is moved in the retraction direction A2 while the positioning section 52 is being gripped, the end section 54B comes into contact with the inclined section 65. By this, pressure is applied to the case 54 in the advance direction A1, and the cover 55 and the base section 56 move in the retraction direction A2 with respect to the case 54.

[0151] In this case, as shown in FIG. 14, the displacement section 55C moves over the locking section 54C toward the retraction direction A2 and moves to the retraction direction A2 side of the locking section 54C. By this, the cutter blade 50 held by the base section 56 is not exposed from the cutter accommodation section 51. In this way, the cutter unit 40 moves from the exposed position P31 to the non-exposed position P32.

[0152] As described above, in accordance with a detachment operation of detaching the cutter unit 40 from the cutter carriage 41, the cutter carriage 41 does not contact the contact section 56C, and the end section 54B of the case 54 comes into contact with the inclined section 65. By this, the displacement section 55C moves over the locking section 54C, and the base section 56 is disposed at the non-exposed position P32. In this case, the cutter unit 40 is in a non-exposed state in which the cutter blade 50 is not exposed from the case 54.

Effects of the Second Embodiment

[0153] Effects of the second embodiment will be described.

[0154] (2-1) The protrusion section 61E and the contact section 56C come into contact with each other in accordance with the attachment operation of attaching the cutter unit 40 to the cutter carriage 41. In this case, the displacement section 55C moves over the locking section 54C and the base section 56 is arranged at the exposed position P31, so that the base section 56 enters the exposed state. By this, the medium 90 can be processed by exposing the cutter blade 50.

[0155] On the other hand, in accordance with the detachment operation of detaching the cutter unit 40 from the cutter carriage 41, the protrusion section 61E and the contact section 56C do not contact each other. In this case, the displacement section 55C moves over the locking section 54C and the base section 56 is disposed at the non-exposed position P32, whereby the non-exposed state is achieved. Therefore, it is possible to improve the safety of the cutter blade 50 when the cutter blade 50 is replaced.

Third Embodiment

[0156] Next, a third embodiment will be described.

[0157] As shown in FIGS. 17 to 21, in the third embodiment, the cutter case 53 includes a first cover 71 and a second cover 72 instead of the cover 55. That is, the cutter unit 40 includes the first cover 71 and the second cover 72.

[0158] The first cover 71 is positioned further in the advance direction A1 than the second cover 72. The second cover 72 includes a positioning section 52. The second cover 72 is fixed to the case 54. The first cover 71 is not fixed to the case 54 and the second cover 72.

[0159] Together with the case 54, the first cover 71 and the second cover 72 accommodate the cutter blade 50 and the biasing section 58. The upper surfaces of the first cover 71 and the second cover 72 are planar. That is, the surfaces of the first cover 71 and the second cover 72 are flat.

[0160] As shown in FIG. 19, the first cover 71 includes a first end section 71A and a first cutout section 71B. The first end section 71A is an end section of the first cover 71 in the retraction direction A2. The first cutout section 71B is provided in the first end section 71A. The first cutout

section **71B** is provided at the first intersecting direction **B1** side of the first cover **71**.

[0161] The second cover **72** includes a second end section **72A** and a second cutout section **72B**. The second end section **72A** is an end section of the second cover **72** in the advance direction **A1**. The second cutout section **72B** is provided in the second end section **72A**. The second cutout section **72B** is provided on the second intersecting direction **B2** side of the second cover **72**.

[0162] The first cover **71** is movable along the advance/retract direction **A**. The first cover **71** is movable between the exposed position **P31** and the non-exposed position **P32**. The exposed position **P31** is a position at which the cutter blade **50** is exposed. The non-exposed position **P32** is a position in which the cutter blade **50** is not exposed.

[0163] The first cover **71** and the second cover **72** are disposed such that the first end section **71A** and the second end section **72A** overlap each other in the intersecting direction **B**. Specifically, the first end section **71A** and the second end section **72A** are disposed so as to overlap each other in the intersecting direction **B** in both of the case in which the first cover **71** is disposed at the exposed position **P31** and the case in which the first cover **71** is disposed at the non-exposed position **P32**. By this, even when the first cover **71** is moved to the exposed position **P31**, the inside of the cutter case **53** is not exposed in the first intersecting direction **B1**. By this, it is possible to avoid that foreign matter enters the interior of the cutter case **53**, and it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

[0164] The case **54** is configured to hold the cutter blade **50**. The cutter blade **50** is fixed to the case **54** from the first intersecting direction **B1** by the fixing member **57**. The case **54** includes an engagement groove **54D**. The engagement groove **54D** is a groove provided on the upper surface of the case **54**. The engagement groove **54D** is engageable with a restricting section **73** (to be described later).

[0165] The cutter unit **40** includes the restricting section **73**, a restricting elastic member **74**, and a biased section **75**. The restricting section **73**, the restricting elastic member **74**, and the biased section **75** are provided to the first cover **71**.

[0166] The restricting section **73** regulates movement of the first cover **71** by engaging with the engagement groove **54D**. The restricting section **73** may be a lock pin. The restricting elastic member **74** is provided between the bottom surface of the first cover **71** and the restricting section **73**. The restricting elastic member **74** biases the restricting section **73** in the second intersecting direction **B2**.

[0167] The restricting section **73** engages with the engagement groove **54D** when the first cover **71** is disposed at the non-exposed position **P32**. The restricting section **73** is disengaged from the engagement groove **54D** when the first cover **71** is disposed at the exposed position **P31**.

[0168] The biased section **75** is provided so as to protrude from the bottom surface of the first cover **71** in the second intersecting direction **B2**. The biased section **75** receives a biasing force from the biasing section **58**. The biasing section **58** biases the first cover **71** in the advance direction **A1** with respect to the case **54** and the second cover **72** via the biased section **75**.

[0169] As shown in FIG. **18**, the first cover **71** includes an engagement section **71C**. The first cover **71** includes a pair of engagement sections **71C**. The pair of engagement sections **71C** are arranged in the widthwise direction **X**. The pair of engagement sections **71C** are configured to protrude outward in the widthwise direction **X** at both widthwise direction **X** end sections of the first cover **71**.

[0170] The cutter carriage **41** includes an engaged section **66**. The cutter carriage **41** includes a pair of engaged sections **66**. The pair of engaged sections **66** are arranged in the widthwise direction **X**. The pair of engaged sections **66** are provided so as to protrude inward in the widthwise direction **X** from respective ones of the pair of unit guide sections **64**. The engaged section **66** is provided on the advance direction **A1** side of the unit holding section **60**. The engaged section **66** is engageable with the engagement section **71C**.

[0171] When the cutter unit **40** moves from the non-exposed position **P32** to the exposed position

P31, the engagement section **71C** engages with the engaged section **66**. By this, as the cutter unit **40** moves in the advance direction **A1**, the first cover **71** moves in the retraction direction **A2** with respect to the case **54** and the second cover **72** against the biasing force of the biasing section **58**.

Operation of the Third Embodiment

[0172] The operation of the third embodiment will be described.

[0173] As shown in FIG. **20**, when the unit fixing section **61** is at the release position **P22**, the attachment operation of attaching the cutter unit **40** to the cutter carriage **41** is performed.

Specifically, when the unit fixing section **61** is at the release position **P22**, the cutter unit **40** is held in the holding region **60A** of the cutter carriage **41** from the retraction direction **A2** side.

[0174] In such a case, as the cutter unit **40** moves in the advance direction **A1** in the holding region **60A**, the engagement section **71C** engages with the engaged section **66**. By this, the first cover **71** moves in the retraction direction **A2** against the biasing force of the biasing section **58**, and the restricting section **73** separates from the engagement groove **54D**. By this, the cutter unit **40** is disposed at the exposed position **P31**.

[0175] As shown in FIG. **21**, the unit fixing section **61** rotates from the release position **P22** to the fixed position **P21**. By this, the cutter unit **40** is fixed to the cutter carriage **41** in a state of being disposed at the exposed position **P31**.

[0176] In this way, the engagement section **71C** engages with the engaged section **66** in accordance with the attachment operation of attaching the cutter unit **40** to the cutter carriage **41**. By this, the first cover **71** moves to the exposed position **P31** against the biasing force of the biasing section **58**, and the cutter blade **50** enters an exposed state of being exposed from the first cover **71**.

[0177] Next, a detachment operation of detaching the cutter unit **40** from the cutter carriage **41** is performed when the unit fixing section **61** is at the fixed position **P21**. Specifically, the unit fixing section **61** rotates from the fixed position **P21** to the release position **P22**. When the unit fixing section **61** is at the release position **P22**, the cutter unit **40** is removed from the cutter carriage **41**.

[0178] In this case, as shown in FIG. **19**, when the positioning section **52** is gripped and the cutter unit **40** is moved in the retraction direction **A2**, the engagement section **71C** of the cutter unit **40** does not engage with the engaged section **66**. By this, the restricting section **73** engages with the engagement groove **54D** by the biasing force of the biasing section **58**, and the first cover **71** moves in the advance direction **A1**. By this, the cutter unit **40** is disposed at the non-exposed position **P32**.

[0179] In this way, in accordance with the detachment operation of detaching from the cutter carriage **41**, the engagement section **71C** does not engage with the engaged section **66**. The cutter unit **40** is such that in such a case the first cover **71** is moved to the non-exposed position **P32** by the biasing force of the biasing section **58**, and the cutter unit **40** enters the non-exposed state in which the cutter blade **50** is not exposed from the first cover **71**.

Effects of the Third Embodiment

[0180] Effects of the third embodiment will be described.

[0181] (3-1) The engagement section **71C** engages with the engaged section **66** in accordance with the attachment operation of attaching the cutter unit **40** to the cutter carriage **41**. In this case, the first cover **71** is moved to the exposed position **P31** against the biasing force of the biasing section **58** to be in the exposed state. By this, the medium **90** can be processed by exposing the cutter blade **50**.

[0182] On the other hand, in accordance with the detachment operation of detaching the cutter unit **40** from the cutter carriage **41**, the engagement section **71C** does not engage with the engaged section **66**. In this case, the first cover **71** is moved to the non-exposed position **P32** by the biasing force of the biasing section **58**, so that the first cover **71** is brought into the non-exposed state. Therefore, it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

[0183] (3-2) The restricting section **73** engages with the engagement groove **54D** by movement of the first cover **71** to the non-exposed position **P32**, and disengages from the engagement groove

54D by movement of the first cover **71** to the exposed position **P31**. According to this configuration, the movement of the first cover **71** can be regulated in the non-exposed position **P32** by the restricting section **73** engaging with the engagement groove **54D**. Therefore, it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

[0184] (3-3) The surface of the first cover **71** is flat. By this, an erroneous operation can be suppressed. Therefore, it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

Fourth Embodiment

[0185] Next, a fourth embodiment will be described.

[0186] As shown in FIG. **22**, in the fourth embodiment, the unit fixing section **61** includes an attachment operation restricting section **61F**. That is, the cutter carriage **41** includes the attachment operation restricting section **61F**. The attachment operation restricting section **61F** is provided so as to protrude from the bottom surface of the main body section **61A** in the second intersecting direction **B2**.

[0187] The attachment operation restricting section **61F** includes a restriction surface **61G**. The restriction surface **61G** is a surface located on the advance direction **A1** side of the attachment operation restricting section **61F**. The restriction surface **61G** is a surface facing the advance direction **A1**.

[0188] When the cutter unit **40** is not properly arranged in the holding region **60A**, the restriction surface **61G** abuts against the cutter unit **40**. By this, the restriction surface **61G** regulates the movement of the unit fixing section **61** to the fixed position **P21**.

[0189] In this manner, when the cutter unit **40** is not positioned in the holding region **60A**, then the attachment operation restricting section **61F** restricts the attachment operation of attaching the cutter unit **40** onto the cutter carriage **41** in response to an attachment operation of attaching the cutter unit **40** onto the cutter carriage **41**.

Effects of the Fourth Embodiment

[0190] Effects of the fourth embodiment will be described.

[0191] (4-1) In a case where the cutter unit **40** is not appropriately disposed in the holding region **60A**, then the attachment operation restricting section **61F** restricts the attachment operation of the cutter unit **40** to the cutter carriage **41** in response to an attachment operation of attaching the cutter unit **40** to the cutter carriage **41**. According to this configuration, if the cutter unit **40** is not properly positioned in the holding region **60A**, the attachment operation of attaching the cutter unit **40** onto the cutter carriage **41** is restricted. By this, erroneous operations can be suppressed. Therefore, it is possible to improve the safety of the cutter blade **50** when the cutter blade **50** is replaced.

Modifications

[0192] The embodiment can be implemented with the following modifications. The embodiments and the following modifications can be implemented in combination with each other as long as there is no technical contradiction. [0193] In the first embodiment and the second embodiment, the contact section **56C** may not be configured to protrude in the first intersecting direction **B1** as long as the contact section **56C** comes into contact with the cutter carriage **41** in accordance with movement of the unit fixing section **61** to the fixed position **P21**. [0194] In the first embodiment and the second embodiment, the protrusion section **61E** may not be provided as long as the configuration is such that the cutter carriage **41** comes into contact with the contact section **56C** in accordance with movement of the unit fixing section **61** to the fixed position **P21**. That is, the cutter carriage **41** may include a portion which comes into contact with the contact section **56C**. [0195] In the second embodiment, the displacement section **55C** may be provided on the base section **56**. The cover **55** and the base section **56** may be integrally provided. [0196] In the second embodiment, the locking section **54C** may lock the case **54** in at least one of the exposed position **P31** and the non-exposed position **P32**. In such a case, the locking section **54C** and the displacement section **55C** may be arbitrarily configured. As a specific example, the locking section **54C** may not protrude in

the first intersecting direction B1, and may have a groove shape. [0197] In the third embodiment, as long as it is a configuration in which the engagement section 71C engages with the engaged section 66 in accordance with the movement of the first cover 71 in the advance/retract direction A, the engagement section 71C and the engaged section 66 may be arbitrarily configured. [0198] In the third embodiment, the restricting section 73 may engage with the engagement groove 54D when the first cover 71 moves to the non-exposed position P32, and may disengage from the engagement groove 54D when the first cover 71 moves to the exposed position P31. [0199] In the third embodiment, the upper surfaces of the first cover 71 and the second cover 72 may not have a planar shape. The upper surface of the first cover 71 and the upper surface of the second cover 72 are at the same surface position, but need not be at the same surface position. [0200] In the third embodiment, the case 54 and the second cover 72 may be integrally provided. The cutter case 53 may include the second cover 72, or may not include the second cover 72. The first cover 71 corresponds to an example of a cover. [0201] In the embodiment described above, the replacement unit accommodation section 39 may be provided at an arbitrary position as long as the user can access the replacement unit accommodation section 39. As a specific example, the replacement unit accommodation section 39 may be provided in the vicinity of a replaceable ink cartridge. The replacement unit accommodation section 39 may be provided in the vicinity of a replaceable waste liquid unit. In this way, it is possible to remove the replacement cutter unit 40 when performing maintenance on the recording device 10. The replacement unit accommodation section 39 may be provided in the accommodation cassette 20. In this way, the replacement cutter unit 40 can be removed when the roll paper 91 is stored in the accommodation cassette 20. The replacement unit accommodation section 39 may be provided at an arbitrary position of the housing 11. The replacement unit accommodation section 39 may be provided on any of the left side surface 14, the right side surface 15, the bottom surface 16, or the top surface 17 of the housing 11. [0202] In the described embodiment, the non-exposed position P32 may be any position, as long as, when viewed from the intersecting direction B, the cutter blade 50 does not protrude from the cutter accommodation section 51 in the advance direction A1. Therefore, the non-exposed position P32 may be a position at which the cutter blade 50 is exposed from the cutter accommodation section 51 or a position at which the cutter blade 50 is not exposed from the cutter accommodation section 51. That is, the exposed position P31 may be a protruding position in which the cutter blade 50 protrudes from the cutter accommodation section 51 in the advance direction A1 when viewed from the intersecting direction B. The non-exposed position P32 may be a non-protruding position at which the cutter blade 50 does not protrude from the cutter accommodation section 51 in the advance direction A1 when viewed from the intersecting direction B. [0203] In the embodiment described above, the attachment operation of attaching the cutter unit 40 on the cutter carriage 41 may be at least one of the operation of placing the cutter unit 40 on the cutter carriage 41 or the operation of moving the unit fixing section 61 to the fixed position P21. [0204] In the above-described embodiment, the detachment operation of detaching the cutter unit 40 from the cutter carriage 41 may be at least one of the operation of moving the unit fixing section 61 to the release position P22 or the operation of taking out the cutter unit 40 from the cutter carriage 41. [0205] A lateral type printer may be adopted as the recording device 10. A lateral type printer is a printer in which the recording carriage 36 is movable in two directions of a main scanning direction and a sub-scanning direction. [0206] The cutter blade 50 is not limited to a rotary blade. The cutter blade 50 may be, for example, a plate-shaped cutter blade, or may be configured to cut the medium 90 by a pair of rotary cutter blades. The cutter unit 40 includes a cutter blade 50 that cuts the medium 90, but this is not a limitation. For example, the cutter unit 40 may include a cutter blade that processes the medium 90 so as to form a crease in the medium 90. As described above, the cutter unit 40 may be a unit having a cutter blade for processing the medium 90 in the transport path 24. That is, recording device 10 may include a processing unit. [0207] The recording device 10 may be configured to record an image on at least one of the roll paper 91 or the paper sheet 92 as the

medium **90**. The medium **90** may be of any type as long as it can be processed by the cutter blade **50** in the recording device **10**. The medium **90** may be a paper sheet, a film or sheet made of a resin, a resin-metal composite film, a laminate film, a woven fabric, a nonwoven fabric, a metal foil, a metal film, a ceramic sheet, clothing, or the like. [0208] The liquid can be arbitrarily selected as long as it can record on the medium **90** by being deposited on the medium **90**. For example, the ink includes an ink in which particles, such as a pigment or metal particles, of a functional material made of a solid material are dissolved, dispersed, or mixed in a solvent, and includes various compositions such as water-based ink, oil-based ink, gel ink, and hot-melt ink. [0209] The recording device **10** is not limited to being an inkjet printer and may be a dot impact printer. The recording method of recording device **10** may be a laser method instead of the ink jet method. [0210] In this specification, the term “at least one of” refers to one or more of the desired alternatives. As an example, if the number of options is two, then the phrase “at least one of” as used herein means only one option or both of the two options. As another example, if the number of options is three or more, then the phrase “at least one of” as used herein means only one option or any combination of two or more options.

Supplementary Notes

[0211] Hereinafter, technical ideas grasped from the above-described embodiment and modifications, and their operations and effects will be described. The present technical concept and its operation effects can be combined with each other within a technically consistent range.

[0212] (A) A recording device includes a transport section that transports a medium in a transport direction along a transport path; a recording section that records an image on the medium transported by the transport section; a cutter unit including a cutter blade for processing the medium in the transport path; and a cutter carriage configured to move the cutter unit in a widthwise direction intersecting the transport direction, wherein the cutter unit is attachable to and detachable from the cutter carriage and, in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, the cutter unit enters an exposed state in which the cutter blade becomes exposed and, in accordance with a detachment operation of detaching the cutter unit from the cutter carriage, enters a non-exposed state in which the cutter blade is not exposed.

[0213] According to this configuration, it is possible to process the medium by exposing the cutter blade in accordance with the attachment operation of attaching the cutter unit to the cutter carriage. On the other hand, in accordance with the detachment operation of the cutter unit from the cutter carriage, the cutter blade is not exposed. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0214] (B) The recording device may be such that the cutter unit includes a base section that holds the cutter blade, a cutter case that accommodates the base section, and a biasing section that biases the base section with respect to the cutter case, in a state in which the base section is accommodated in the cutter case, the base section is movable between an exposed position in which the cutter blade is exposed and a non-exposed position in which the cutter blade is not exposed, the base section includes a contact section that, in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, comes into contact with the cutter carriage, and the cutter unit is such that when the cutter carriage and the contact section come into contact with each other in accordance with the attachment operation to the cutter carriage, the base section is moved against biasing force from the biasing section into the exposed position and the cutter blade enters the exposed state of being exposed from the cutter case and when, in accordance with the detachment operation from the cutter carriage, the cutter carriage and the contact section do not contact each other, the base section is moved by the biasing force of the biasing section to the non-exposed position and the cutter blade enters the non-exposed state in which the cutter blade is not exposed from the cutter case.

[0215] According to this configuration, the cutter carriage and the contact section come into contact with each other in accordance with the attachment operation of attaching the cutter unit to the cutter

carriage. In this case, the exposed state is achieved by moving the base section to the exposed position against the biasing force of the biasing section. By this, it is possible to process the medium by exposing the cutter blade.

[0216] On the other hand, in accordance with the detachment operation of detaching the cutter unit from the cutter carriage, the cutter carriage and the contact section do not contact each other. In this case, the base section is moved to the non-exposed position by the biasing force of the biasing section, so that the non-exposed state is achieved. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0217] (C) The recording device may be such that the contact section does not protrude from a surface of the cutter unit while the cutter unit is being mounted on the cutter carriage.

[0218] According to this configuration, when the cutter unit is mounted on the cutter carriage, the contact section does not protrude from the surface of the cutter unit. By this, erroneous operations can be suppressed. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0219] (D) The cutter unit may include a base section that holds the cutter blade, a case that accommodates the base section, and a displacement section that is displaced in conjunction with the base section, in a state of being accommodated in the case, the base section is movable between an exposed position in which the cutter blade is exposed and a non-exposed position in which the cutter blade is not exposed, the base section includes a contact section that comes into contact with the cutter carriage in accordance with the attachment operation of attaching the cutter unit to the cutter carriage, the case includes a locking section that locks with the displacement section in at least one of the exposed position and the non-exposed position, and the cutter unit is such that when the cutter carriage and the contact section come into contact with each other in accordance with the attachment operation to the cutter carriage, the displacement section moves over the locking section so that the base section is disposed at the exposed position and also the cutter blade enters the exposed state of being exposed from the case and in accordance with the detachment operation of detaching from the cutter carriage, the displacement section moves over the locking section without the cutter carriage and the contact section contacting each other, so that the base section is positioned at the non-exposed position and also the cutter blade enters the non-exposed state in which the cutter blade is not exposed from the case.

[0220] According to this configuration, the cutter carriage and the contact section come into contact with each other in accordance with the attachment operation of attaching the cutter unit to the cutter carriage. In this case, the displacement section moves over the locking section and the base section is disposed at the exposed position, whereby the base section enters the exposed state. By this, it is possible to process the medium by exposing the cutter blade.

[0221] On the other hand, in accordance with the detachment operation of detaching the cutter unit from the cutter carriage, the cutter carriage and the contact section do not contact each other. In this case, the displacement section moves over the locking section and the base section is disposed at the non-exposed position, whereby the non-exposed state is achieved. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0222] (E) The recording device may be such that the cutter unit includes a case that accommodates the cutter blade, a cover that, together with the case, accommodates the cutter blade, and a biasing section configured to bias the cover against the case, the cover is movable between an exposed position in which the cutter blade is exposed and a non-exposed position in which the cutter blade is not exposed, the cover has an engagement section, the cutter carriage has an engaged section with which the engagement section is engageable, and the cutter unit is such that in accordance with the attachment operation of attaching to the cutter carriage, the engagement section engages with the engaged section, the cover is moved to the exposed position against the biasing force of the biasing section, and the cutter blade enters the exposed state of being exposed from the cover and in accordance with a detachment operation of detaching from the cutter carriage, the

engagement section does not engage with the engaged section, the cover is moved to the non-exposed position by the biasing force of the biasing section, and the cutter blade enters the non-exposed state in which the cutter blade is not exposed from the cover.

[0223] According to this configuration, the engagement section engages with the engaged section in accordance with the attachment operation of the cutter unit to the cutter carriage, and thus the cover moves to the exposed position against the biasing force of the biasing section to enter the exposed state. By this, it is possible to process the medium by exposing the cutter blade.

[0224] On the other hand, the engagement section does not engage with the engaged section in accordance with the detachment operation of detaching the cutter unit from the cutter carriage. In this case, the cover is moved to the non-exposed position by the biasing force of the biasing section, so that the cover is brought into the non-exposed state. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0225] (F) The recording device may be such that the cover includes a restricting section that restricts movement of the cover, the case includes an engagement groove engageable with the restricting section, and the restricting section engages with the engagement groove when the cover moves to the non-exposed position and disengages from the engagement groove when the cover moves to the exposed position.

[0226] According to this configuration, in the non-exposed position, the movement of the cover can be restricted by the restricting section engaging with the engagement groove. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0227] (G) The recording device may be such that a surface of the cover is planar.

[0228] According to this configuration, the surface of the cover is planar. By this, erroneous operations can be suppressed. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0229] (H) The recording device may be such that the cutter carriage includes an attachment operation restricting section that, when the cutter unit is not disposed in a holding region, restricts the attachment operation of attaching the cutter unit to the cutter carriage in accordance with the attachment operation of the cutter unit on the cutter carriage.

[0230] According to this configuration, if the cutter unit is not positioned in the holding region, the attachment operation of attaching the cutter unit to the cutter carriage is restricted. By this, erroneous operations can be suppressed. Therefore, it is possible to improve the safety of the cutter blade when the cutter blade is replaced.

[0231] (I) The recording device may further include a housing and a replacement unit accommodation section provided in the housing and configured to accommodate a replacement cutter unit.

[0232] According to this configuration, it is possible to easily replace the replacement cutter unit by taking out the cutter unit from the replacement unit accommodation section.

Claims

1. A recording device comprising: a transport section that transports a medium in a transport direction along a transport path; a recording section that records an image on the medium transported by the transport section; a cutter unit including a cutter blade for processing the medium in the transport path; and a cutter carriage configured to move the cutter unit in a widthwise direction intersecting the transport direction, wherein the cutter unit is attachable to and detachable from the cutter carriage and, in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, the cutter unit enters an exposed state in which the cutter blade becomes exposed and, in accordance with a detachment operation of detaching the cutter unit from the cutter carriage, enters a non-exposed state in which the cutter blade is not exposed.
2. The recording device according to claim 1, wherein the cutter unit includes a base section that

holds the cutter blade, a cutter case that accommodates the base section, and a biasing section that biases the base section with respect to the cutter case, in a state in which the base section is accommodated in the cutter case, the base section is movable between an exposed position in which the cutter blade is exposed and a non-exposed position in which the cutter blade is not exposed, the base section includes a contact section that, in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, comes into contact with the cutter carriage, and the cutter unit is such that when the cutter carriage and the contact section come into contact with each other in accordance with the attachment operation to the cutter carriage, the base section is moved against biasing force from the biasing section into the exposed position and the cutter blade enters the exposed state of being exposed from the cutter case and when, in accordance with the detachment operation from the cutter carriage, the cutter carriage and the contact section do not contact each other, the base section is moved by the biasing force of the biasing section to the non-exposed position and the cutter blade enters the non-exposed state in which the cutter blade is not exposed from the cutter case.

3. The recording device according to claim 2, wherein the contact section does not protrude from a surface of the cutter unit while the cutter unit is being mounted on the cutter carriage.

4. The recording device according to claim 1, wherein the cutter unit includes a base section that holds the cutter blade, a case that accommodates the base section, and a displacement section that is displaced in conjunction with the base section, in a state of being accommodated in the case, the base section is movable between an exposed position in which the cutter blade is exposed and a non-exposed position in which the cutter blade is not exposed, the base section includes a contact section that comes into contact with the cutter carriage in accordance with the attachment operation of attaching the cutter unit to the cutter carriage, the case includes a locking section that locks with the displacement section in at least one of the exposed position and the non-exposed position, and the cutter unit is such that when the cutter carriage and the contact section come into contact with each other in accordance with the attachment operation to the cutter carriage, the displacement section moves over the locking section so that the base section is disposed at the exposed position and also the cutter blade enters the exposed state of being exposed from the case and in accordance with the detachment operation of detaching from the cutter carriage, the displacement section moves over the locking section without the cutter carriage and the contact section contacting each other, so that the base section is positioned at the non-exposed position and also the cutter blade enters the non-exposed state in which the cutter blade is not exposed from the case.

5. The recording device according to claim 1, wherein the cutter unit includes a case that accommodates the cutter blade, a cover that, together with the case, accommodates the cutter blade, and a biasing section configured to bias the cover against the case, the cover is movable between an exposed position in which the cutter blade is exposed and a non-exposed position in which the cutter blade is not exposed, the cover has an engagement section, the cutter carriage has an engaged section with which the engagement section is engageable, and the cutter unit is such that in accordance with the attachment operation of attaching to the cutter carriage, the engagement section engages with the engaged section, the cover is moved to the exposed position against the biasing force of the biasing section, and the cutter blade enters the exposed state of being exposed from the cover and in accordance with a detachment operation of detaching from the cutter carriage, the engagement section does not engage with the engaged section, the cover is moved to the non-exposed position by the biasing force of the biasing section, and the cutter blade enters the non-exposed state in which the cutter blade is not exposed from the cover.

6. The recording device according to claim 5, wherein the cover includes a restricting section that restricts movement of the cover, the case includes an engagement groove engageable with the restricting section, and the restricting section engages with the engagement groove when the cover moves to the non-exposed position and disengages from the engagement groove when the cover moves to the exposed position.

7. The recording device according to claim 5, wherein a surface of the cover is planar.

8. The recording device according to claim 1, wherein the cutter carriage includes an attachment operation restricting section that, when the cutter unit is not disposed in a holding region, restricts the attachment operation of attaching the cutter unit to the cutter carriage in accordance with the attachment operation of the cutter unit on the cutter carriage.

9. The recording device according to claim 1, further comprising: a housing and a replacement unit accommodation section provided in the housing and configured to accommodate a replacement cutter unit.

10. A cutter unit attachable to and detachable from a cutter carriage with respect to a recording device, the recording device including a transport section that transports a medium in a transport direction along a transport path, a recording section that records an image on the medium transported by the transport section, and the cutter carriage, which moves the cutter unit in a widthwise direction intersecting the transport direction, the cutter unit comprising: a cutter blade for processing the medium in the transport path, wherein the cutter unit enters an exposed state in which the cutter blade becomes exposed in accordance with an attachment operation of attaching the cutter unit to the cutter carriage, and enters a non-exposed state in which the cutter blade is not exposed in accordance with a detachment operation of detaching the cutter unit from the cutter carriage.
